

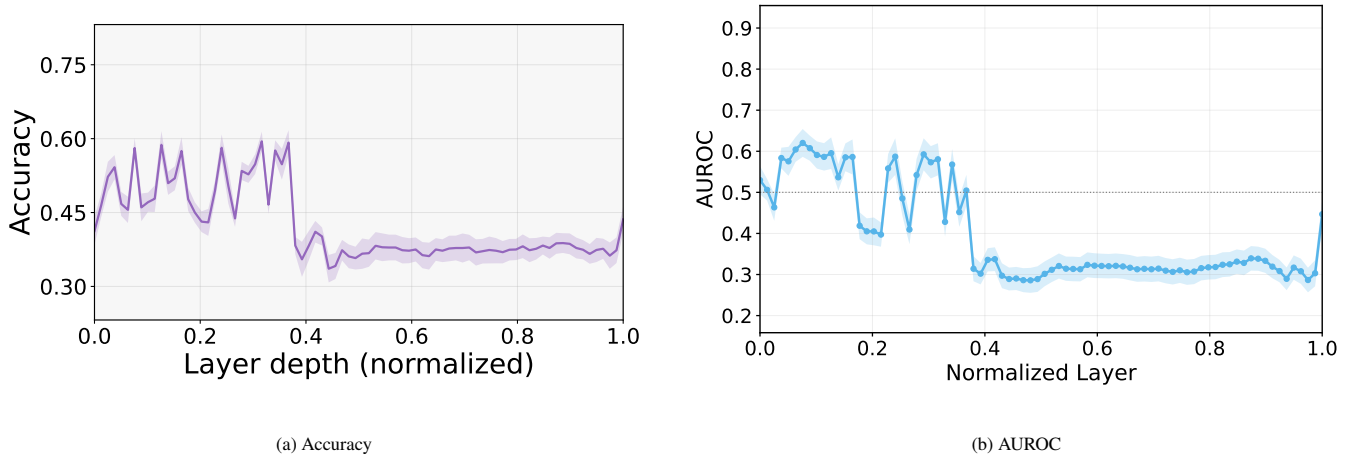


Figure 12: Layerwise probe classification of evaluation versus deployment context for Nemotron3-49B, using the mean-token representation. Nemotron3-49B’s last-token performance is confined to the first few layers and near chance thereafter; under mean-token pooling, early layers separate the contexts weakly in the probe-aligned direction (AUROC ≈ 0.55 – 0.6), while middle and late layers separate them in the direction anti-correlated with the probe (AUROC ≈ 0.3). Accuracy in (a) uses a threshold calibrated on GSM8K/TruthfulQA vs. WildChat; its below-chance values at middle and late layers reflect this sign reversal, not an absence of separation. Normalized layer depth denotes the layer index divided by the total number of hidden-state layers returned by the model.

continuous. Its main limitation is that it can only detect monotonic dependence: if the true relationship between s_ℓ and J is non-monotonic (e.g., judge scores are elevated for both extreme positive and extreme negative probe scores), Spearman will report a value close to zero even when the two variables are strongly dependent. We compute ρ layerwise for each model, separately for prompt-last and response-last readouts.

Mutual information. The mutual information $I(s_\ell; J)$ between two random variables measures how much knowing one of them reduces uncertainty about the other:

$$\begin{aligned}
 I(s_\ell; J) &= \mathbb{E}_{p(s_\ell, J)} \left[\log \frac{p(s_\ell, J)}{p(s_\ell)p(J)} \right] \\
 &= D_{\text{KL}}(p(s_\ell, J) \parallel p(s_\ell)p(J)).
 \end{aligned}
 \tag{G.1}$$

Equivalently, $I(s_\ell; J) = H(J) - H(J \mid s_\ell)$, where H denotes Shannon entropy. We report MI in nats (natural-log units). MI is zero if and only if s_ℓ and J are statistically independent, and is non-negative otherwise; larger values indicate stronger dependence. Unlike Spearman, MI captures arbitrary statistical dependencies, including nonlinear, non-monotonic, and threshold-like relationships, which is important here because the verbalization process need not be a smooth monotonic function of the probe score.

We estimate MI using the Kraskov k -nearest-neighbor estimator (Kraskov, Stögbauer, and Grassberger 2004) in its mixed continuous–discrete formulation, as implemented in scikit-learn. This estimator is non-parametric, avoids binning, and infers entropies from distances to the k -th nearest neighbor in the joint (s_ℓ, J) space. We treat both s_ℓ and J as continuous variables for estimator compatibility (with small jitter to break ties); k is set to the scikit-learn default. For each layer we pool all candidate prompts, drop pairs with missing values, and require at least three valid samples and non-zero variance in both variables before estimating MI. The two statistics are complementary: Spearman tests for monotonic association, while MI measures whether the probe score reduces uncertainty about the judge score in any form. A model could in principle have $\rho \approx 0$ but $I > 0$ (non-monotonic dependence), or $\rho \neq 0$ but I near baseline (a weak monotonic trend that does not substantially reduce uncertainty). Reporting both distinguishes these cases; since both $|\rho|$ and MI are small in our results, the per-prompt coupling is genuinely weak rather than merely non-monotonic.

Random-direction control. For both statistics, we report a random-direction baseline: for each model and layer we sample a unit-norm direction uniformly at random in representation space, recompute $s_\ell^{\text{rand}}(x)$, and recompute ρ and I against the same judge scores. Because MI estimators have non-zero bias on finite samples and Spearman has finite-sample variance, the random baseline gives us an empirical null: any genuine signal in the trained probe should rise above the random direction’s value at the same model, layer, and sample size. We average the random baseline over multiple seeds and report the maximum across seeds as a conservative upper baseline (seeds in Appendix A.1).

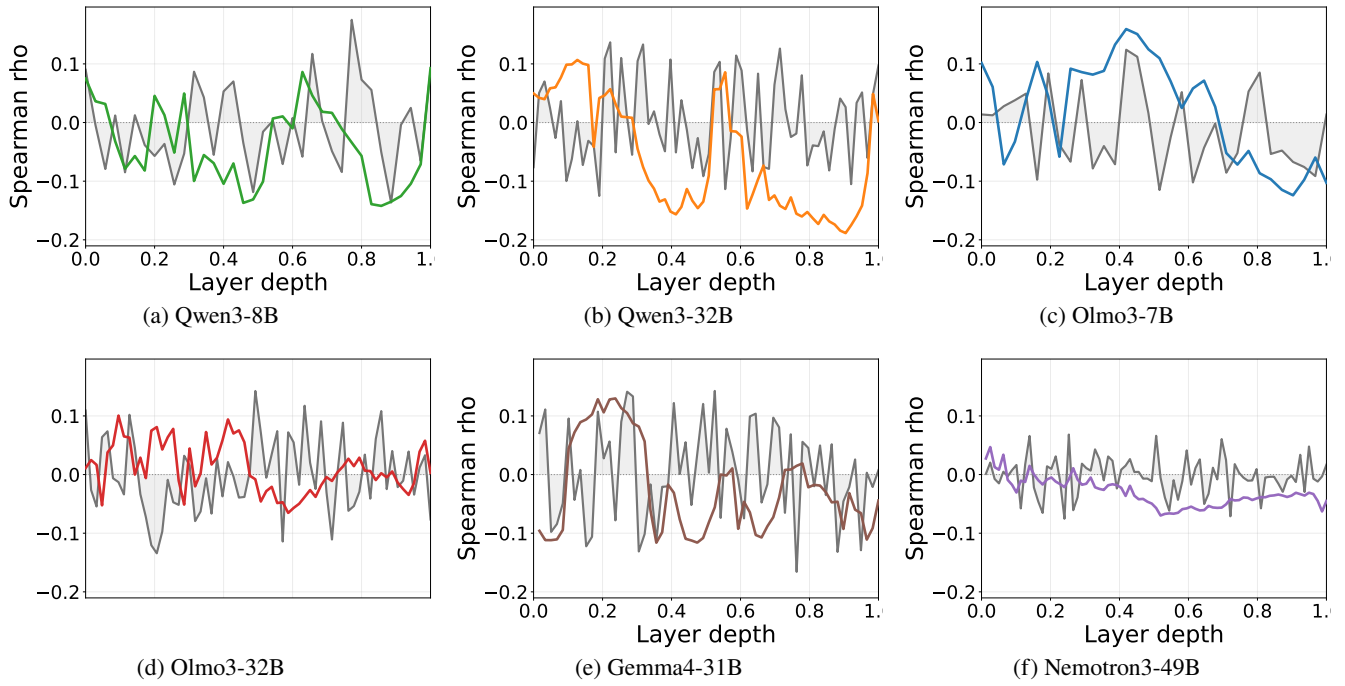


Figure 13: Spearman correlation between the last token of prompt’s probe score and judge scores across the six models. Gray bands show the random-direction baseline. Peak $|\rho_\ell| < 0.19$ for every model. Normalized layer (normalized layer depth) denotes the layer index divided by the total number of hidden-state layers returned by the model.

Correlations between the last token of prompt’s score and the judge score. Figures 13 and 14 report Spearman correlation and mutual information, respectively, between the prompt-last probe score and the judge score across all six models. Both statistics remain close to the random-direction baseline (gray bands) at every layer, with peak $|\rho_\ell| < 0.19$ and peak $I(s_\ell; J) < 0.04$ nats across all models. The agreement between the two statistics indicates that the weak coupling we observe is not only non-monotonicity but reflects genuinely low dependence between the probe score and verbalization.

Correlation between response score and judge score In this section, we report the correlation between the response score and the judge score across all evaluated models. Response score is short for response’s last token score. We present results using both Spearman correlation (Figure 15) and mutual information (Figure 16). Unlike the prompt-based analysis above, here the representations are extracted from the last token of the model’s response rather than the prompt. Even at the response readout, where one might expect a tighter coupling because the judge directly reads the same generation, both statistics remain close to the random-direction baseline.

G.3 Judge Robustness Analysis

As in Appendix D, repeating the coupling analysis of Appendix G with J^{100} in place of J^3 does not change the qualitative conclusion. For the four reasoning checkpoints, Table 8 reports the peak across all layers of $|\rho_\ell|$ and the Kraskov mutual information $I(s_\ell; J^{100})$ for each probe readout. The peak Spearman correlation never exceeds $|\rho_\ell| = 0.17$ and the peak MI never exceeds 0.15 nats; on the `</think>`-token readout for Qwen (the only family where the close-tag is detectable in this MASK run), peak $|\rho_\ell| \leq 0.09$ and peak MI ≤ 0.013 nats. Across these readouts the learned probe $\hat{v}_\ell$ sits essentially on the random-direction baseline at every layer (Figure 17); i.e., the learned probe has no meaningful advantage over a random direction in predicting J^{100} prompt-by-prompt. The J^3 and J^{100} judges agree on this: both keep the per-prompt probe–judge coupling near random across every readout, consistent with the dissociation reported in Section 3 and Appendix G.

G.4 Direct Comparison to the AUROC Result of Section 3

On the same six models and the same prompts, peak AUROC > 0.70 (Section 3) coexists with peak $|\rho_\ell| < 0.19$ and peak $I(s_\ell; J) < 0.04$ nats. The probe direction reliably separates two prompt sources in hidden states yet predicts $J(x)$ only weakly at the prompt-wise level. These two measurements are not redundant: AUROC contrasts two prompt sample-types (evaluation vs deployment) at the population level, while ρ and I ask whether higher probe scores predict higher judge scores prompt-by-prompt within a fixed pool. If the probe direction captured the signal that *drives* verbalization, then within evaluation-sourced prompts,

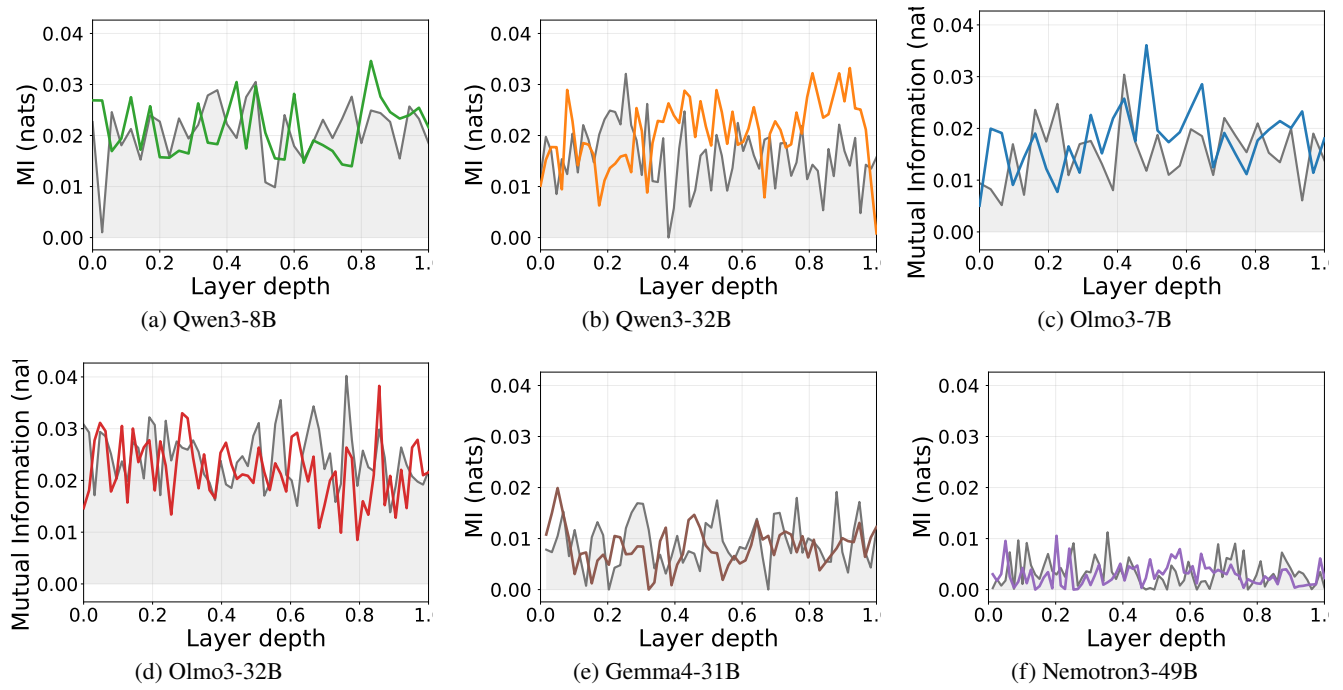


Figure 14: Mutual information between the last token of prompt’s probe score and judge scores across the six models. Gray bands show the random-direction baseline. Peak $I(s_\ell; J) < 0.04$ nats for every model. Layer depth (normalized layer depth) denotes the layer index divided by the total number of hidden-state layers returned by the model.

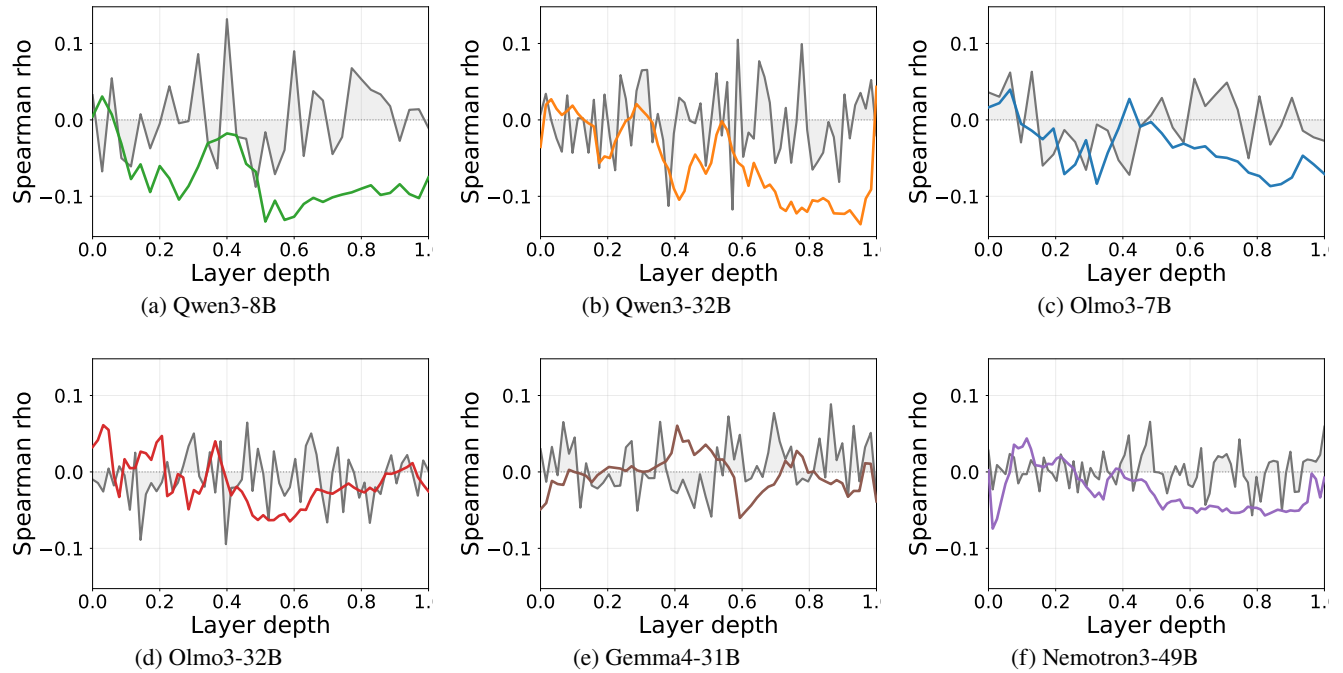


Figure 15: Spearman correlation between response-last probe score and judge score across the six models. Layer depth (normalized layer depth) denotes the layer index divided by the total number of hidden-state layers returned by the model.